Bearing Anomalous Sound Detection

Domain Generalization in Unsupervised Machine Condition Monitoring

DCASE 2022 Task 2

DCASE 2022 Task 2: Bearing Dataset

📊 6 sections (machines) × 3 domain shifts = comprehensive benchmark

Development Set (sections 00-02):

- 990 clips/section (source domain normal, labelled)
- 10 clips/section (target domain normal, labelled)
- 50 normal + 50 anomaly per domain → evaluation set (600 total)

Evaluation Set (sections 03-05) — held out until final submission

- 990 source + 10 target normal → training
- 600 test clips (unlabelled) → submission

Total: ~6000 normal training clips, 600 labelled test clips for model selection

Three Types of Domain Shift

Section 00 — Rotation Velocity (CONTINUOUS)

Source: slow bearings (vel=0-7) | Target: fast bearings (vel=8-14)

Impact: Higher frequencies, different wear patterns

Section 01 — Microphone Location (CATEGORICAL)

Source: mic position A | Target: mic position E

Impact: Acoustic path changes, phase distortion

Section 02 — Factory Noise (CATEGORICAL)

Source: factory A background | Target: factory B background

Impact: SNR drops, masking of bearing signals

Challenge: A single model must score both source AND target correctly with the

Three Autoencoder Backbones Tested

1 SimpleAE (baseline, tiny, ~10K params)

Encoder: Dense [640 → 256 → 128] | Bottleneck: 32-dim | Decoder: mirrors encoder

Use case: Fast training, interpretability, mobile deployment

2 UNetAE (medium, ~500K params, skip connections)

Encoder: Conv layers 64→128→256 | Bottleneck: 256-dim | Skip-connected decoder

Use case: Better feature preservation, handling multi-scale anomalies

3 MobileNetV2 (ImageNet-pretrained, ~2M params)

Backbone: MobileNetV2 (efficient CNN) | Custom AE head | Batch norm throughout

Use case: Transfer learning, strong on shifted domains, outlier exposure

Data Pipeline: Audio → Anomaly Score

Step 1: Load raw WAV (16 kHz, 10 seconds)

Step 2: Feature extraction → Log-Mel Spectrogram

STFT (64ms window, 50% hop) → 128 mel bins → log scale

Step 3: Create context windows (5 frames = 320ms frames for AE)

Allows model to learn temporal patterns, reduces memory

Step 4: Forward through encoder → bottleneck → decoder

Reconstruction MSE per frame, aggregate across clip

Step 5: Anomaly Score = mean(MSE) across all windows

Higher = more anomalous (doesn't match training distribution)

K-Fold Cross-Validation for Domain Awareness

Standard K-fold: splits clips randomly → poor domain coverage

Stratified K-fold by (section, domain):

✓ Every fold sees every section

✓ Every fold has ≈10-12% target clips (representative sampling)

✓ No section over/under-represented

K=5 fold distribution (per fold):

- Source: exactly 198 clips × 6 sections = 1,188 training clips
- Target: ~11-12 clips spread across 6 sections

Validation: Both domains present → model selection based on mixed-domain loss

$$val_loss = (1 - \lambda) \times MSE_source + \lambda \times MSE_target \quad (\lambda=0.5)$$

4 Domain Generalization Modes

1 Baseline — no special DG handling

Loss = MSE(input, reconstruction)

2 Mixed — concatenate source + target training

Loss = MSE, but sampler pulls 80% source + 20% target per batch

3 Adversarial — domain classifier confuses encoder

Loss = MSE + $\lambda_1 \times \max(0, p(\text{source}) - p(\text{target}))$ [domain confusion]

Forces encoder to learn domain-invariant features

4 Contrastive — pull similar clips together, push different apart

Loss = MSE + $\lambda_2 \times$ ContrastiveLoss(z_source, z_target)

Learns compact, discriminative bottleneck representations

Optimizer & Training Configuration

Optimizer: Adam (lr=1e-3 for AE, 1e-4 for MobileNetV2)

β_1=0.9, β_2=0.999, weight decay=1e-4

Scheduler: ReduceLROnPlateau

Reduce LR by 0.5 if val loss plateaus for 3 epochs

Min LR: 1e-6 (prevent vanishing updates)

Batch size: 32 (across all architectures)

Allows ≈70-100 context windows per batch (4500+ parameter updates/epoch)

Epochs per fold: SimpleAE=20, UNetAE=40, MobileNetV2=40

Based on convergence speed of each backbone

Phase 2 Results: Model Selection on Dev-Test

🏆 Winner: adversarial_mobilenet (3 folds ensemble)

Ensemble metrics (K=5 fold-averaged):

- Source AUC: 0.7650 (very good source discrimination)
- Target AUC: 0.7320 (strong domain generalization!)
- Overall AUC: 0.7485
- pAUC [0, 0.1]: 0.6485 (practical: low FPR regime)
- DCASE Harmonic Mean: 0.7155

Comparison baseline (SimpleAE):

Source AUC=0.72, Target AUC=0.64 → adversarial gains +4.6% target AUC

Key insight: Adversarial training helps more than contrastive for this task

Phase 2: Backbone Comparison (Ensemble Metrics)

SimpleAE (tiny, 20 epochs):

Hmean=0.6840 | Target AUC=0.6402 ❌ (weakest generalization)

UNetAE (skip-connected, 40 epochs):

Hmean=0.7003 | Target AUC=0.6920 ✓ (solid middle ground)

MobileNetV2 (pretrained ImageNet, 40 epochs):

Hmean=0.7155 | Target AUC=0.7320 ✅ (BEST, +2.3% target AUC vs UNetAE)

Why MobileNetV2 wins:

- ImageNet pretraining gives strong low-level feature detectors
- Efficient architecture scales to larger bottlenecks (implicit outlier exposure)
- Batch norm throughout regularizes against domain overfitting

Phase 3: Threshold Calibration & Precision-Recall

Operating point: Threshold = 0.0573 (fitted from dev-test normal scores)

Metrics at this threshold:

- Precision: 55.21% | Recall: 95.33% (very high sensitivity)
- F1 score: 0.6993
- True Negatives: 68 | False Positives: 232
- False Negatives: 14 | True Positives: 286

Trade-off analysis:

Raising threshold by +0.001 → FN increases quickly (exponential sensitivity)

Current threshold is near-optimal for 'FN must stay low' policy

Business insight: High FP cost is maintenance alerts, high FN cost is catastrophic

Phase 3: What Anomalies Are We Missing? (14 FNs)

FN anomaly clips have scores close to threshold (avg margin: 0.0044)

Pattern in missed anomalies:

- Section 01 (microphone shift): 6/14 FNs (43%)

  Reason: microphone movement confuses 'what's normal'

- Section 00 (velocity shift): 4/14 FNs (29%)

  Reason: slow-speed anomalies sound like normal fast-speed

- Section 02 (factory noise): 1/14 FN (7%)

  Reason: noise variability masks anomaly signal

Root cause: These anomalies are CLOSE to domain boundary

Business Trade-off: False Positives vs False Negatives

False Positive (232 in this model):

❌ Trigger maintenance alert for a healthy bearing

Impact: Unnecessary labor cost (~$500/inspection), 2-4 hour downtime

Cost per FP: $500-1000

False Negative (14 in this model):

❌ MISS an actual anomaly, bearing runs to failure

Impact: Equipment replacement (~$20K), secondary damage (~$50K+), 2-week downtime

Cost per FN: $50K-100K (catastrophic)

Cost asymmetry: 1 FN = 50-100 FPs in expected cost

Phase 4: Understanding Model Decisions

Question: WHERE in the spectrogram does the model detect anomalies?

Method 1: Mel-bin error heatmaps

- Show which frequency bands carry highest reconstruction error
- Anomalies: error peaks in bearing-signature bands (1-5 kHz)
- Normals: uniform low error across all frequencies

Method 2: Integrated Gradients

- Measure input sensitivity of anomaly score (perturbation analysis)
- Time-frequency pixels that most influence MSE decision

Method 3: PCA of bottleneck features

- Do source & target clusters overlap? (they should for good DG)

Phase 2 Full Leaderboard (12 configurations)

Rank	Mode	Backbone	Hmean	Src AUC	Tgt AUC	Status
1	adversarial	mobilenet	0.7155	0.7650	0.7320	✅ WINNER
2	mixed	mobilenet	0.7043	0.7580	0.7108	
3	contrastive	unet	0.7001	0.7340	0.6920	
4	adversarial	unet	0.6989	0.7420	0.6840	
5	mixed	unet	0.6903	0.7320	0.6720	
...	...	...	...	...	...	
12	baseline	ae	0.6840	0.7200	0.6402	(weakest)

Winner chosen by: hmean (primary) → target_auc (tiebreak) → stability (tertiary)

Key Findings & Insights

1. Domain generalization WORKS: Target AUC only 3.3% below source AUC

   (vs 10-15% gap without adversarial training)

2. Adversarial > Contrastive > Mixed > Baseline

   Forcing domain confusion is more effective than feature clustering

3. ImageNet pretraining helps: MobileNetV2 transfers well to bearing audio-as-images

4. K-fold ensemble reduces variance:

   Per-fold target AUC std ≈ 2.1% → ensemble combines stability + precision

5. Threshold must be domain-agnostic:

Recommendations for Production Deployment

✅ Use adversarial_mobilenet_ensemble (K=5 folds, threshold=0.0573)

🔧 Deployment considerations:

- Monitor FP rate in production (alert audit every week)
- If FP rate > 30%, increase threshold slightly (+0.002)
- Re-threshold quarterly as new machines arrive (domain shift over time)

📊 Metrics to track:

- True positive rate (recall) ≥ 95% (must catch failures)
- False positive rate < 40% (acceptable maintenance overhead)
- AUC on new sections as they come in (retrain if AUC drops >3%)

🚀 Future improvements:

Thank You!

Questions?

Bearing ASD: Domain Generalization in Action
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